CSE-1223
Discrete Mathematics
Segment 6: Counting

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Basic Counting Principle

The Basics of Counting

- **A counting problem:**

Each user on a computer system has a password, which is six to eight characters long, where each characters is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

- **This section introduces:**
  - a variety of other counting problems
  - the basic techniques of counting

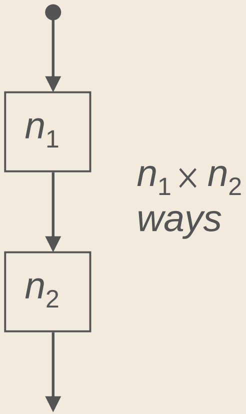
The Basics of Counting

- We first present two basic counting principles, the **product rule** and the **sum rule.**
- Then we will show how they can be used to solve many different counting problems

The Basics of Counting

The product rule:

Suppose that a procedure can be broken down into two tasks. If there are n1 ways to do the first task and n2 ways to do the second task after the first task has been done, then there are n1 n2 ways to do the procedure.



The Basics of Counting

Example 1: A new company with just two employees, Sanchez and Patel, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?

Solution: The procedure of assigning offices to these two employees consists of assigning an office to Sanchez, which can be done in 12 ways, then assigning an office to Patel different from the office assigned to Sanchez, which can be done in 11 ways. By the product rule, there are $12 \cdot 11 = 132$ ways to assign offices to these two employees.

The Basics of Counting

Example 2: The chairs of an auditorium are to be labeled with an uppercase English letter followed by a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently?

Solution: $\underset{\text{letter}}{\underline{26}} \times \underset{\boxed{\begin{array}{l} 1 \leq x \leq 100 \\ x \in \boldsymbol{N} \end{array}}}{\underline{100}} = 2600$ ways to label chairs.

The Basics of Counting

Example 3: How many different bit strings of length seven are there?

Solution:

$$\begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \square & \square & \square & \square & \square & \square & \square \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ 0,1 & 0,1 & 0,1 & & \ldots\ldots & & 0,1 \end{array} \quad \rightarrow \; 2^7 \text{ ways}$$

The Basics of Counting

Example 4: How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits. [**Spr'23**]

Solution:

$$\underbrace{\square\ \square\ \square}_{\text{letter}} \qquad \underbrace{\square\ \square\ \square}_{\text{digit}} \rightarrow 26^3 \cdot 10^3$$

The Basics of Counting

Example 5: How many different license plates can be made using either two or three letters followed by 5 digits? [**Spr’17**]

Solution:

$$\underbrace{\square\,\square}_{\text{letter}} \text{ either } \underbrace{\square\,\square\,\square}_{\text{letter}} \,.\, \underbrace{\square\,\square\,\square\,\square\,\square}_{\text{digit}} \rightarrow (26^2+26^3).10^5$$

The Basics of Counting

Example 6: The DNA code is made up of four letters: A, C, G, and T. How many 5-element DNA sequences

i) end with A?

ii) start with T and end with G?

iii) contain only A and T?

iv) do not contain C? ? [**Aut'23**]

Solution:

The Basics of Counting

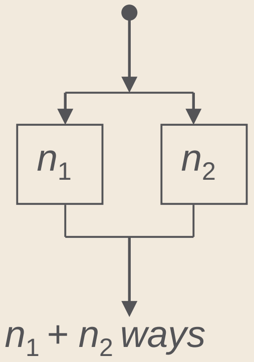
Example 6: The DNA code is made up of four letters: A, C, G, and T. How many 5-element DNA sequences
i) end with A?
ii) start with T and end with G?
iii) contain only A and T?
iv) do not contain C? [**Aut'23**]
Solution:

i) A. end with A?	ii) start with T and end with G?	iii) contain only A and T?	iv) do not contain C?
1 Element : 4 ways (A, G , C, T) 2 Element : 4 ways (A, G , C, T) 3 Element : 4 ways (A, G , C, T) 4 Element : 4 ways (A, G , C, T) 5 Element : 1 way (since it has to be A) Using Product Rule: = 4 x 4 x 4 x 4 x 1 = 256 sequences 256 sequences end with A.	1 Element : 1 way (since it has to be T) 2 Element : 4 ways (A, G , C, T) 3 Element : 4 ways (A, G , C, T) 4 Element : 4 ways (A, G , C, T) 5 Element : 1 way (since it has to be G) Using Product Rule: = 1 x 4 x 4 x 4 x 1 = 64 sequences 64 sequences start with T and end with G.	1 Element : 2 ways (since it has to be A and T only) 2 Element : 2 ways (since it has to be A and T only) 3 Element : 2 ways (since it has to be A and T only) 4 Element : 2 ways (since it has to be A and T only) 5 Element : 2 ways (since it has to be A and T only) Using Product Rule: = 2 x 2 x 2 x 2 x 2 = 32 sequences 32 sequences contain only A and T only.	1 Element : 3 ways (since it cannot be C) 2 Element : 3 ways (since it cannot be C) 3 Element : 3 ways (since it cannot be C) 4 Element : 3 ways (since it cannot be C) 5 Element : 3 ways (since it cannot be C) Using Product Rule: = 3 x 3 x 3 x 3 x 3 = 243 sequences 243 sequences do not contain C.

The Basics of Counting

The sum rule:

If a first task can be done in n1 ways and a second task in n2 ways, and if these tasks cannot be done at the same time then there are n1+n2 ways to do either task.



The Basics of Counting

Example 1: Suppose that either a member of the mathematics faculty or a student who is a mathematics major is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student?

Solution: There are 37 ways to choose a member of the mathematics faculty and there are 83 ways to choose a student who is a mathematics major. Choosing a member of the mathematics faculty is never the same as choosing a student who is a mathematics major because no one is both a faculty member and a student.

By the sum rule it follows that there are $37 + 83 = 120$ possible ways to pick this representative.

The Basics of Counting

Example 2: A student can choose a computer project from one of three lists. The three lists contain 23, 15 and 19 possible projects respectively. How many possible projects are there to choose from?

Solution: $23+15+19=57$ projects.

The Basics of Counting

Example 3: In a version of the computer language BASIC, the name of a variable is a string of one or two alphanumeric characters, where uppercase and lowercase letters are not distinguished. (An alphanumeric character is either one of the 26 English letters or one of the 10 digits.) Moreover, a variable name must begin with a letter and must be different from the five strings of two characters that are reserved for programming use. How many different variable names are there in this version of BASIC?

Solution: Let V equal the number of different variable names in this version of BASIC. Let V_1 be the number of these that are one character long and V_2 be the number of these that are two characters long. Then by the sum rule, $V = V_1 + V_2$. Note that $V_1 = 26$, because a one-character variable name must be a letter. Furthermore, by the product rule, there are $26 \cdot 36$ strings of length two that begin with a letter and end with an alphanumeric character. However, five of these are excluded, so $V_2 = 26 \cdot 36 - 5 = 931$. Hence, there are $V = V_1 + V_2 = 26 + 931 = 957$ different names for variables in this version of BASIC.

The Basics of Counting

Exercise: A multiple-choice test contains 10 questions. There are four possible answers for each question.

a) In how many ways can a student answer the questions on the test if the student answers every question?

b) In how many ways can a student answer the questions on the test if the student can leave answers blank? [**Spr'18**]

The Basics of Counting

The Division Rule:

There are n/d ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w, exactly d of the n ways correspond to way w.

The Basics of Counting

Example 1: Suppose that an automated system has been developed that counts the legs of cows in a pasture. Suppose that this system has determined that in a farmer's pasture there are exactly 572 legs. How many cows are there is this pasture, assuming that each cow has four legs and that there are no other animals present?

Solution:

Let n be the number of cow legs counted in a pasture. Because each cow has four legs, by the division rule we know that the pasture contains $n/4$ cows. Consequently, the pasture with

572 cow legs has $572/4 = 143$ cows in it.

The Basics of Counting

Example 2: On each of the 22 work days in a particular month, every employee of a start-up venture was sent a company communication. If a total of 4642 total company communications were sent, how many employees does the company have, assuming that no staffing changes were made that month? [**Aut'23**]

Solution:

Let n be the number of company communication. Since every employee receives one communication on each of the 22 work days in the month, by the division rule we know that the number of communications sent each day is 4642/22 = 211.

So, the company has 211 employees.

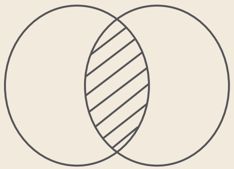
2

Inclusion-Exclusion Principle

Inclusion-Exclusion Principle

Principle: If a task can be done in either n1 ways or n2 ways, then the number of ways to do the task is n1 + n2 minus the number of ways to do the task that are common to the two different ways. [**Spr'18, Aut'17, Spr'16**]

The principle of **inclusion–exclusion** is also known as the **subtraction rule**.



A B

$$\left|A \cup B\right| = \left|A\right| + \left|B\right| - \left|A \cap B\right|$$

Inclusion-Exclusion Principle

Example 1: How many bit strings of length eight either start with a 1 bit or end with the two bits 00 ? [**Spr'23**]

Solution:

1	2	3	4	5	6	7	8
□	□	□	□	□	□	□	□
	↑	↑	. . . .				

① 1 0,1 0,1 → A total of 2^7 bit strings

② . . . . 0 0 → A total of 2^6 bit strings

③ 1 . . . . 0 0 → A total of 2^5 bit strings

$\therefore\ 2^7 + 2^6 - 2^5$ bit strings

$1\ \underbrace{\_\ \_\ \_\ \_\ \_\ \_\ \_}_{2^7 = 128 \text{ ways}}$

$\underbrace{\_\ \_\ \_\ \_\ \_\ \_}_{2^6 = 64 \text{ ways}}\ 0\ 0$

$1\ \underbrace{\_\ \_\ \_\ \_\ \_}_{2^5 = 32 \text{ ways}}\ 0\ 0$

The Basics of Counting

Example 2: How many bit strings of length 5 either starts with a 1 bit or end with two bits 00? [**Spr'17**]

Solution: Use the subtraction rule.

- Number of bit strings of length 5 that start with a 1 bit: $2^4 = 16$
- Number of bit strings of length 5 that end with bits 00: $2^3 = 8$
- Number of bit strings of length 5 that start with a 1 bit and end with bits 00 : $2^2 = 4$

Hence, the number is 16 + 8 – 4 = 20.

$1\,\underbrace{\_\,\_\,\_\,\_}$

$2^4 = 16$ ways

$\underbrace{\_\,\_\,\_}\,0\,0$

$2^3 = 8$ ways

$1\,\underbrace{\_\,\_}\,0\,0$

$2^2 = 4$ ways

Home Work: How many bit strings of length 10 either starts with three 0s or end with two 0s? [**Spr'18**]

The Basics of Counting

Example 2: How many bit strings of length 5 either starts with a 1 bit or end with two bits 00? [**Spr'17**]

Solution: Use the subtraction rule.

- ➢ Number of bit strings of length 5 that start with a 1 bit: $2^4 = 16$
- ➢ Number of bit strings of length 5 that end with bits 00: $2^3 = 8$
- ➢ Number of bit strings of length 5 that start with a 1 bit and end with bits 00 : $2^2 = 4$

Hence, the number is $16 + 8 - 4 = 20$.

$\underline{1}\ \underbrace{\_\ \_\ \_\ \_}$

$2^4 = 16$ ways

$\underbrace{\_\ \_\ \_}\ \underline{0}\ \underline{0}$

$2^3 = 8$ ways

$\underline{1}\ \underbrace{\_\ \_}\ \underline{0}\ \underline{0}$

$2^2 = 4$ ways

Home Work: How many bit strings of length 10 either starts with three 0s or end with two 0s? [**Spr'18**]

Answer: 352

3

Pigeonhole Principle

Pigeonhole Principle

Principle: If k is a positive integer and k + 1 or more objects are placed into k boxes, then there is at least one box containing two or more of the objects. [**Aut’16, Spr’17, Aut’17, Spr’23**]

Proof: We prove the pigeonhole principle using a proof by contraposition. Suppose that none of the k boxes contains more than one object. Then the total number of objects would be at most k.
This is a contradiction, because there are at least k + 1 objects

Pigeonhole Principle

Example 1: Among any 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.

Example 2: In any group of 27 English words, there must be at least two that begin with the same letter.

Example 3: How many students must be in a class to guarantee that at least two students receive the same score on the final exam ? (0~100 points)

Solution: 102. (101+1)

Pigeonhole Principle

Example 4: Explain how the pigeonhole principle can be used to show that among any 11 integers, at least two must have the same last digit. [**Aut’17**]

Solution:

Unit digit of all integers are one of the following 10 integers.

0, 1, 2, 3, 4, 5, 6, 7,8 and 9

If 11 integers are randomly chosen, then using pigeonhole principle there will be at least two integers with the same unit digit.

Generalized Pigeonhole Principle

Principle: If N objects are placed into k boxes, then there is at least one box containing at least ⌈N/k⌉ objects. [**Spr'16,18**]

Proof: We prove the pigeonhole principle using a proof by contraposition. Suppose that none of the k boxes contains more than one object. Then the total number of objects would be at most k.
This is a contradiction, because there are at least k + 1 objects

Generalized Pigeonhole Principle

Example 1: Among 100 people there are at least $\left\lceil \frac{100}{12} \right\rceil = 9$ who were born in the same month. (100 objects, 12 boxes).

Example 2: There are 50 baskets of apples. Each basket contains at least one apple and no more than 24 apples. Show that there are 3 baskets containing the same number of apples. [**Spr'16**]

Solution: Consider each of the baskets as pigeons. So, n=50.

According to how many apples are in the baskets, each basket is placed in one of the 24 pigeonholes. So, k=24.

Calculate the ratio of pigeons (n) to pigeonholes (k).

$[n/k] = [50/24] = 3$

Generalized Pigeonhole Principle

Example 3: What is the minimum number of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five possible grades, A, B, C, D and F.

Solution: To use pigeonhole principle, first find boxes and objects.

- Suppose that for each grade, we have a box that contains students who got that grade.
- The number of boxes is 5, by the generalized pigeonhole principle, to have at least 6 ($\lceil N/5 \rceil$) students at the same box, the total number of the students must be at least N = 5 . 5 + 1 = 26.

Generalized Pigeonhole Principle

Example 4: What is the minimum number of students required in a discrete mathematics class to be sure that at least eight will receive the same grade, if there are five possible grades, A+, A, A- and F. [**Spr'23**]

Solution: To use pigeonhole principle, first find boxes and objects.

- ❑ Suppose that for each grade, we have a box that contains students who got that grade.
- ❑ The number of boxes is 4, by the generalized pigeonhole principle, to have at least 8 ($\lceil N/4 \rceil$) students at the same box, the total number of the students must be at least N = 7 . 4 + 1 = 29.

Generalized Pigeonhole Principle

Example 5: Show that among any group of five (not necessarily consecutive) integers, there are two with the same remainder when divided by 4. [**Spr'18**]

Solution:

8 = 0 + 2* 4, reminder 0

1 = 1 + 0 * 4, reminder 1

10 = 2 + 2 * 4,reminder 2

15 = 3+ 3 * 4,reminder 3

We have chosen 0, 1, 2,3 which is all of the n's that will satisfy: $0 \le n < 4$.

Any number we choose next will certainly have to have an n value of either 0, 1, 2 or 3

Generalized Pigeonhole Principle

Example 6: What is the minimum number of students, each of whom comes from one of the 50 states, who must be enrolled in a university to guarantee that there are at least 100 who come from the same state? [**Aut’17**]

Solution: The generalized pigeonhole principle applies here. The pigeons are the students (no slur intended), and the pigeonholes are the states, 50 in number. By the generalized pigeonhole principle if we want there to be at least 100 pigeons in at least one of the pigeonholes, then we need to have a total of N pigeons so that IN/501 ~ 100. This will be the case as long as N ~ 99 · 50 + 1 = 4951. Therefore we need at least 4951 students to guarantee that at least 100 come from a single state

4

Permutation and Combination

Permutation

A **permutation** is an arrangement of some elements in which order matters. In other words a Permutation is an ordered Combination of elements.

The number of permutations of ‘n’ different things taken ‘r’ at a time is denoted by ^nP_r

$^nP_r = n!/(n-r)!$

where $n! = 1.2.3....(n-1).n$

Permutation

Example 1: We have to form a permutation of three digit numbers from a set of numbers S={1,2,3}. Different three digit numbers will be formed when we arrange the digits. The permutation will be = 123, 132, 213, 231, 312, 321

Permutation

Example 2: In how many ways can we select three students from a group of five students to stand in line for a picture? In how many ways can we arrange all five of these students in a line for a picture?

Solution:

First, note that the order in which we select the students matters. There are five ways to select the first student to stand at the start of the line. Once this student has been selected, there are four ways to select the second student in the line. After the first and second students have been selected, there are three ways to select the third student in the line. By the product rule, there are $\mathbf{5 \cdot 4 \cdot 3 = 60}$ ways to select three students from a group of five students to stand in line for a picture.

To arrange all five students in a line for a picture, we select the first student in five ways, the second in four ways, the third in three ways, the fourth in two ways, and the fifth in one way. Consequently, there are $\mathbf{5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120}$ ways to arrange all five students in a line for a picture.

Permutation

Example 3: How many ways are there to select a first-prize winner, a second-prize winner, and a third-prize winner from 100 different people who have entered a contest?

Solution:

Because it matters which person wins which prize, the number of ways to pick the three prize winners is the number of ordered selections of three elements from a set of 100 elements, that is, the number of 3-permutations of a set of 100 elements. Consequently, the answer is $\mathbf{P(100, 3) = 100 \cdot 99 \cdot 98 = 970{,}200}$

Permutation

Example 4: Suppose that there are eight runners in a race. The winner receives a gold medal, the second-place finisher receives a silver medal, and the third-place finisher receives a bronze medal. How many different ways are there to award these medals, if all possible outcomes of the race can occur and there are no ties?

Solution:

The number of different ways to award the medals is the number of 3-permutations of a set with eight elements. Hence, there are $P(8, 3) = 8 \cdot 7 \cdot 6 = 336$ possible ways to award the medals.

Permutation

Example 5: Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?

Solution:

The number of possible paths between the cities is the number of permutations of seven elements, because the first city is determined, but the remaining seven can be ordered arbitrarily. Consequently, there are $7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$ ways for the saleswoman to choose her tour. If, for instance, the saleswoman wishes to find the path between the cities with minimum distance, and she computes the total distance for each possible path, she must consider a total of 5040 paths!

Permutation

Example 6 : How many ways are there for 3 penguins and 6 puffins to stand in a line so that
i) all puffins stand together?
ii) all penguins stand together? [**Aut'23**]

Solution:

i) Let's consider the 6 puffins as a block. Now arrange the 3 penguins and block of puffins, which can be done in ${}^4P_4 = 4!$ ways. Then we arrange the 6 puffins within their block. So there are a total of $4! \cdot 6! = 17,280$ arrangements.

ii) Let's consider the 3 penguins as a block. Now arrange the 6 puffins and block of penguins, which can be done in ${}^7P_7 = 7!$ ways. Then we arrange the 3 penguins within their block. So there are a total of $7! \cdot 3! = 30,240$ arrangements.

Combination

A **combination** is selection of some given elements in which order does not matter.

The number of all combinations of n things, taken r at a time is –

$$^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$$

Combination

Example 1: Find the number of subsets of the set {1,2,3,4,5,6} having 3 elements.

Solution:

The cardinality of the set is 6 and we have to choose 3 elements from the set. Here, the ordering does not matter. Hence, the number of subsets will be ${}^6C_3 = \frac{6!}{3!(6-3)!} = \frac{6!}{3!3!} = \frac{6*5*4*3*2*1}{6*6} = 20$.

Combination

Example 2: There are 6 men and 5 women in a room. In how many ways we can choose 3 men and 2 women from the room?

Solution:

The number of ways to choose 3 men from 6 men is $^6C_3$ and the number of ways to choose 2 women from 5 women is $^5C_2$. Hence, the total number of ways is –

$^6C_3 \times {}^5C_2$

$= \frac{6!}{3!3!} \times \frac{5!}{2!3!}$

$= 20 \times 10$

$=200$

Combination

Example 3: How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school?

Solution:

The answer is given by the number of 5 combinations of a set with 10 elements.

The number of such combinations is

$^{10}C_5$

$= \frac{10!}{5!(10-5)!}$

$= \frac{10!}{5!5!}$

= 252 ways

Combination

Example 4: A group of 30 people have been trained to form a cricket team. How many ways are there to select a team of 11 players to form a best 11 squad. [**Spr'17, Aut'16**]

Solution:

Selecting 11 players from the group = ${}^{30}C_{11}$ = 5,46,27,300 ways

Home Work: suppose an ice-cream shop has three different types of ice-cream. How many different ways can seven ice-cream be chosen. [**Spr'18**]

Combination

Example 5: Suppose that there are 9 faculty members in the mathematics department and 11 in the computer science department. How many ways are there to select a committee to develop a discrete mathematics course at a school if the committee is to consist of three faculty members from the mathematics department and four from the computer science department? [**Spr'18**]

Solution:

Selecting three faculty members from the mathematics department = ${}^{9}C_{3}$

Selecting four faculty members from the computer department= ${}^{11}C_{4}$

Therefore from the product rule= = ${}^{9}C_{3} * {}^{11}C_{4} = 84 * 330 = 27{,}720$ ways

Combination

Example 6: A coin is flipped 8 times where each flip comes up either heads or tails. How many possible outcomes

i. are there in total?

ii. contain exactly two heads?

iii. contain at most three tails?

Solution:

i) Tossing each coin gives two possible outcomes: Heads, or tails

A coin is flipped eight times.

The possible outcomes for tossing each coin is 2.

Total number of possible outcomes = $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^8 = 256$

Combination

Example 6: A coin is flipped 8 times where each flip comes up either heads or tails. How many possible outcomes

i. are there in total?

ii. contain exactly two heads?

iii. contain at most three tails?

Solution:

ii) 8 coins are tossed.

Need to calculate the possible ways to get exactly 2 heads.

Use the combination formula to calculate the possible outcomes that contain exactly two heads.

$$^{n}C_{r} = \frac{n!}{r!(n-r)!}$$

$$^{8}C_{2} = \frac{8!}{2!(8-2)!} = \frac{8!}{2!6!} = \frac{40320}{2 * 720} = 28$$

Combination

Example 6: A coin is flipped 8 times where each flip comes up either heads or tails. How many possible outcomes

i. are there in total?

ii. contain exactly two heads?

iii. contain at most three tails?

Solution:

iii) 8 coins are tossed.

To find the number of possible outcomes that contains at most three tails, use the formula of combination.

Number of possible outcomes $= 0 \text{ tails} + 1 \text{ tail} + 2 \text{ tails} + 3 \text{ tails}$

$= C(8,0) + C(8,1) + C(8,2) + C(8,3)$

$= \frac{8!}{0!8!} + \frac{8!}{1!7!} + \frac{8!}{2!6!} + \frac{8!}{3!5!}$

$= 1 + 8 + 28 + 56$

$= 93$

Combination

Example 7: The English alphabet contains 21 consonants and five vowels. How many strings of six lowercase letters of the English alphabet contain **[Sp’24]**

i. exactly one vowel?

ii. exactly two vowels?

iii. at least one vowel?

Solution:

i. There are 26 letters, of which 21 are consonants and 5 are vowels.

In strings contains 6 letters:

Position vowel: 6 ways

Vowel: 5 ways

Second letter: 21 ways (needs to be a consonant)

Third letter: 21 ways (needs to be a consonant)

Fourth letter: 21 ways (needs to be a consonant)

Fifth letter: 21 ways (needs to be a consonant)

Sixth letter:21 ways (needs to be a consonant)

Use the product rule: $6 * 5 * 21 * 21 * 21 * 21 * 21 = 6 * 5 * 21^5 = 122{,}523{,}030$

Thus, there are 122,523,030 strings containing exactly one vowel.

Combination

Example 7: The English alphabet contains 21 consonants and five vowels. How many strings of six lowercase letters of the English alphabet contain **[Sp'24]**

i. exactly one vowel?

ii. exactly two vowels?

iii. at least one vowel?

Solution:

ii. There are 26 letters, of which 21 are consonants and 5 are vowels.

In strings contains 6 letters:

Position vowel: $C(6,2) = \frac{6!}{2!4!} = 15$ ways

1^{st} Vowel: 5 ways

2^{nd} Vowel: 5 ways

Third letter: 21 ways (needs to be a consonant)

Fourth letter: 21 ways (needs to be a consonant)

Fifth letter: 21 ways (needs to be a consonant)

Sixth letter: 21 ways (needs to be a consonant)

Use the product rule: $15 * 5 * 5 * 21 * 21 * 21 * 21 = 15 * 5 * 5 * 21^4 = 72{,}930{,}375$

Thus, there are 72,930,375 strings containing exactly two vowels.

Combination

Example 7: The English alphabet contains 21 consonants and five vowels. How many strings of six lowercase letters of the English alphabet contain **[Sp'24]**

i. exactly one vowel?

ii. exactly two vowels?

iii. at least one vowel?

Solution:

iii. There are 26 letters in a total of which 21 are consonants and 5 are vowels. We are interested in strings contains 6 letters.

There are 26 possible letters for each letter in the string. By the product rule:

Number of strings = $26*26*26*26*26*26=26^6=308,915,776$

When the string contains no vowels, then there are 21 possible letters for each letter in the string. By the product rule:

Number of strings with no vowels = $21*21*21*21*21*21=21^6=85,766,121$

Number of strings with at least one vowel = Number of strings — Number of strings with no vowels $= 26^6 - 21^6 = 308,915,776-85,766,121=223,149,655$

5

Binomial Coefficients

Binomial Coefficients

The **binomial coefficient** is used to denote the number of possible ways to choose a subset of objects of a given numerosity from a larger set.
It is so called because it can be used to write the coefficients of the expansion of a power of a binomial.

The binomial coefficient is **denoted** by
$$\binom{n}{k}$$
and it is read as "n choose k" or "n over k".

Binomial Coefficients

Binomial Theorem:

Let x,y be variables, and let n be a positive integer, then

$$(x+y)^n = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \ldots + \binom{n}{n-1}xy^{n-1} + \binom{n}{n}y^n = \sum_{j=0}^{n}\binom{n}{j}x^{n-j}y^j$$

Binomial Coefficients

Example: What is the expansion of $(x + y)^4$?

Solution: From the binomial theorem it follows that

$$\begin{aligned}(x+y)^4 &= \sum_{j=0}^{4} \binom{4}{j} x^{4-j} y^j \\ &= \binom{4}{0}x^4 + \binom{4}{1}x^3y + \binom{4}{2}x^2y^2 + \binom{4}{3}xy^3 + \binom{4}{4}y^4 \\ &= x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4.\end{aligned}$$

Binomial Coefficients

EXAMPLE 3 What is the coefficient of $x^{12}y^{13}$ in the expansion of $(x+y)^{25}$?

Solution: From the binomial theorem it follows that this coefficient is

$$\binom{25}{13} = \frac{25!}{13!\,12!} = 5{,}200{,}300.$$

EXAMPLE 4 What is the coefficient of $x^{12}y^{13}$ in the expansion of $(2x-3y)^{25}$?

Solution: First, note that this expression equals $(2x+(-3y))^{25}$. By the binomial theorem, we have

$$(2x+(-3y))^{25} = \sum_{j=0}^{25}\binom{25}{j}(2x)^{25-j}(-3y)^j.$$

Consequently, the coefficient of $x^{12}y^{13}$ in the expansion is obtained when $j=13$, namely,

$$\binom{25}{13}2^{12}(-3)^{13} = -\frac{25!}{13!\,12!}2^{12}3^{13}.$$

We can prove some useful identities using the binomial theorem, as Corollaries 1, 2, and 3 demonstrate.

6

Recurrence relations

Recurrence relations

A **recurrence relation** is an equation that recursively defines a sequence where the next term is a function of the previous terms (Expressing Fn as some combination of Fi with i<n).

Example – Fibonacci series – Fn=Fn−1+Fn−2, Tower of Hanoi – Fn=2Fn−1+1

$$f(n) = a^k f(1) + \sum_{j=0}^{k-1} a^j g(n/b^j).$$

Recurrence relations

EXAMPLE 6 Let $f(n) = 5f(n/2) + 3$ and $f(1) = 7$. Find $f(2^k)$, where k is a positive integer. Also, estimate $f(n)$ if f is an increasing function.

Extra Examples

Solution: From the proof of Theorem 1, with $a = 5$, $b = 2$, and $c = 3$, we see that if $n = 2^k$, then

$$\begin{aligned} f(n) &= a^k[f(1) + c/(a-1)] + [-c/(a-1)] \\ &= 5^k[7 + (3/4)] - 3/4 \\ &= 5^k(31/4) - 3/4. \end{aligned}$$

Also, if $f(n)$ is increasing, Theorem 1 shows that $f(n)$ is $O(n^{\log_b a}) = O(n^{\log 5})$. ◀